

SUBJECT : PROBLEM SOLVING AND TESTING USING JAVA

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Task 3 (Easy): Maximum Profit Analyzer

Problem Description

Given daily profit/loss values, find the maximum possible profit obtainable from a contiguous sequence of days using Kadane's Algorithm.

Input Format

- First line contains integer N.
- Second line contains N integers.

Output Format

Print maximum subarray sum.

Constraints

- $1 \leq N \leq 10^5$

Sample Input

8

-2 -3 4 -1 -2 1 5 -3

Sample Output

7

Program:

```
import java.util.*;
public class ProfitAnalyzer {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int currentSum = arr[0];
        int maxSum = arr[0];

        for (int i = 1; i < n; i++) {

            currentSum = Math.max(arr[i], currentSum + arr[i]);

            maxSum = Math.max(maxSum, currentSum);
        }

        System.out.println(maxSum);
    }
}
```

Output:

```
● apple@Nandinis-Macbook-Air PST % javac ProfitAnalyzer.java
● apple@Nandinis-Macbook-Air PST % java ProfitAnalyzer.java
8
-2
3
4
5
-1
8
-6
-7
19
```